Computer programming E140

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September 8, 2025

What will be covered in this course?

- Software: R (and RStudio)
- Mostly basic things in these programs
- Course will not cover many things
- Course will not be deep

Goals

At the end of this course. . .

- . . . you understand basic concepts of R
- . . . you can do a basic analysis in R

Prerequisites

- Basic demographic knowledge (e.g., you know what a ‘rate’ is)
- Basic statistical knowledge (e.g., you know what a ‘mean’ is)
- First experince using statistical software (Stata, Excel, SAS, SPSS, R, . . .)

Already an expert?

- Write a function that takes any integer as input and returns TRUE if the integer is a prime number, otherwise it returns FALSE
- Write a function that takes any integer as input and returns its amicable number if it exists, otherwise it returns FALSE
- Check one of the many tasks on rosettacode.org solved with R
- Reproduce or replicate one of the many reproducible/replicable articles available at Demographic Research
- Work on your own analysis

Materials

Materials will be available from GitHub:

- https://github.com/christiandudel/EDSD2026

Contact

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- Website: http://www.christiandudel.com

Things I work on/I am interested in

- *Topics*: Labor markets, fertility, health
- *Concepts*: Stratification, inequality, life courses, aging
- *Methods*: Longitudinal data analysis, causal inference, (partial) identification, survey methodology

Course schedule

- September 8 (Tue), 09:30 - 11:30: Introduction
- September 8 (Tue), 13:30 - 15:30: R basics
- September 9 (Wed), 09:30 - 11:30: Descriptive statistics
- September 9 (Wed), 13:30 - 15:30: Data visualization
- September 11 (Fri), 13:30 - 15:30: Data handling
- October 5 (Mon), 13:30 - 15:30: Exercise
- October 6 (Tue), 13:30 - 15:30: Data analysis example
- October 7 (Wed), 13:30 - 15:30: Programming (1)
- October 8 (Thu), 9:30 - 11:30: Programming (2)
- October 9 (Fri), 13:30 - 15:30: Programming example

Types of session

- Regular session (6)
- Example session (2)
- Exercise session (1)
- Intro session (1)

Regular sessions

- I show things
- You bring your laptop and follow
- Always possible to ask questions!

Example sessions

- Real applications
- Combine material from previous sessions

Exercise session

- For some sessions, there will be voluntary exercises
- Exercises and solutions are available online (GitHub)
- Exercise session (October 5) provides space to do exercises with my support
- These voluntary exercises have to be distinguished from the (mandatory!) assignment

Assignment: Overview

- One mandatory assignment
- Assignment handed out on September 11
- Deadline: October 18
- Assignment will consist of several tasks

Assignment: Your solutions

- You submit R code as solutions (via email)
- R code should be commented, explaining what is happening
- Code should work without errors

Assignment: Deadline

- Deadline assignment: October 18, 12am/midnight/24:00 (CEST/Berlin time)
- Send your solutions to me (dudel@demogr.mpg.de)
- You will get a confirmation (might take a bit, sorry)
- I might get back to you if I have problems with your file(s)
- It is your responsibility that your files are working!

Assignment: Groups

- You can work in groups
- Actually, I strongly suggest you work in groups!
- Please not more than five people per group
- Please submit your solutions only once per group
- Make clear who is member of the group when submitting

Assignment: Grading

- You can either “pass” or “fail”
- Your code should...
- ...work without errors
- ...be well-documented: Comments!
- ...should be (somewhat) efficient. If one step can do the work then don't use two or more!

Assignment: Summary

- One assignment consisting of coding tasks
- You submit code as solutions, via email
- Deadline: October 18
- You can work in groups
- Pass/fail

Other dates

- September 10 (Thu): 30th anniversary of the MPIDR
- September 16 (Wed), 16:00: MPIDR Summer Party
- October 20 (Tue): EDSD Social Demography Day
- Drinks?

Today: Introduction

- Schedule and formalities
- R and R installation
- Various basic topics:
  - What is programming?
  - Open science and reproducible research
  - Data protection
  - OS, file structure, directories
  - Versions

What is R?

- R is an open source statistical programming language
- First release in 1995
- Used for data analysis and statistical programming

Why use R?

- Free, open source
- Can easily be extended
- More than 22,600 packages available on CRAN
- Many methods are already implemented in R
- Commonly used in both science and industry, top ten programming language
- Many R-related materials: Books, journals, conferences, forums, tutorials...

Why use RStudio?

- R is the programming language
- RStudio is a tool to use R more efficiently (IDE)
- Features:
  - Syntax highlighting, code folding
  - Project management (e.g., GitHub)
  - Markdown support
  - ...

Disclaimer

- R is not the only statistical software and it is fine if you prefer something else
- RStudio is not the only IDE/editor for R (ESS, RKWward, Tinn-R, ...)
- R can be used in many different ways
- Example: base R vs tidyverse vs data.table vs specialized packages
- I do things in certain ways, and this course will follow that
- This does not mean that the solutions from this course are the only or the best way to do things

What do you need to get started?

- R: https://cloud.r-project.org/
- R-Studio: https://www.rstudio.com/

What is programming?

Computer programming is defined as the process of writing a set of instructions called program that directs the computer to perform a specific task.

What is a programming language?

A programming language is an engineered language for expressing computer programs, typically allowing to be written in a human readable manner.

Example:

```
#include <stdio.h>
int main() {
   printf("Hello, World!");
   return 0;
}
```

There are many different programming languages and different types of programming languages.

Programming terminology

- Code: The instructions/text you write
- Run or execute code: Make the computer follow the instructions
- Comment: Part of the code which is not executed
- ...

What is statistical programming?

Statistical programming is the process of using computer programming languages to analyze and manipulate data for statistical purposes.

R is an object-oriented statistical programming language which can be used for statistics, math, demography, and more.

Why statistical programming for EDSD students?

- Statistical programming is a skill on its own, separate from theoretical statistics
- Statistical programming is becoming increasingly important as requirements change (e.g., reproducible research)

Open science

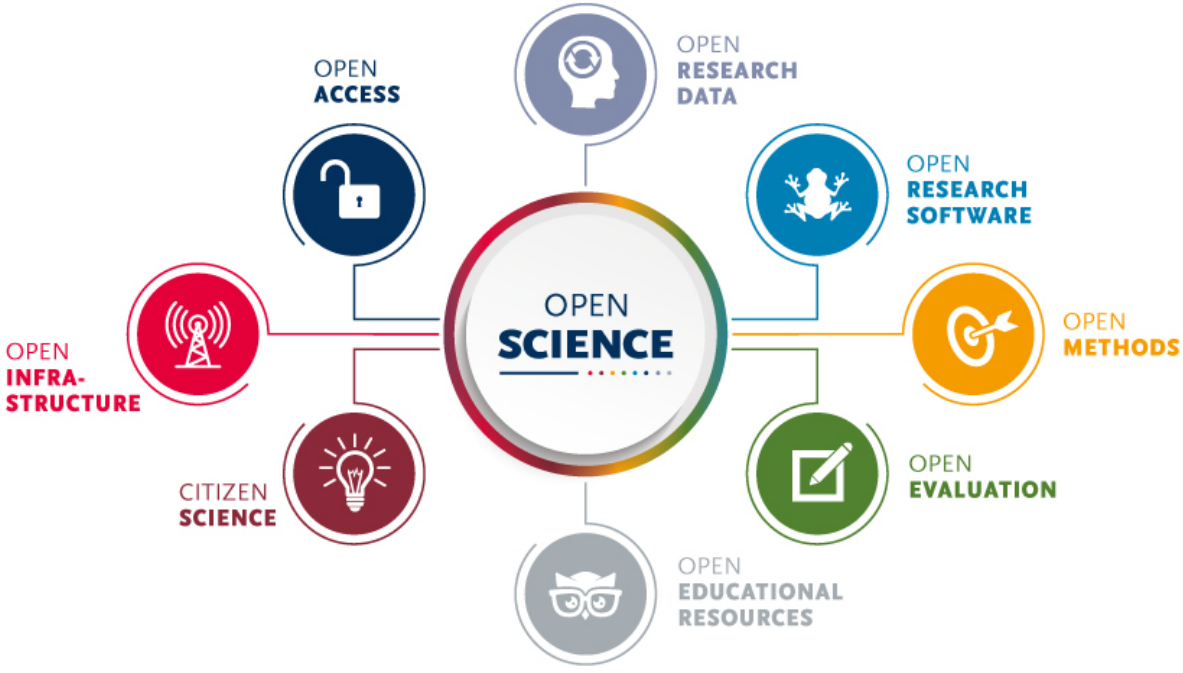


Figure 1: University of Potsdam | ZIM 2023 (CC BY 4.0)

Reproducible research

Research is reproducible if others can generate exactly the same results by using the same data and code. This requires that software, data, and code are openly available.

How can R help?

- R is open source, everyone can use it
- R code can easily be shared (GitHub, Zenodo, journal website. . .)
- R code can include comments which explain what happens
- Data availability might be a limitation

Data protection

- Data protection: protection of personal information
- General Data Protection Regulation (European Union)
  - Since 2018
  - Strict rules, high fines
- Country-specific rules, e.g., Bundesdatenschutzgesetz in Germany
- Data-set specific rules: data use agreements

Data protection in practice

- Make sure to carefully study the use agreement for a specific data set
- Follow these rules!
- Think twice where you store data, also applies to intermediate results
- Do not share data if you are not 100% sure that you are allowed to share it

OS, file structure, directories



Operating systems

- Operating system = core software which manages your hardware and software
- Popular OS: Windows, Linux, MacOS
- Many different versions of each (and sometimes variants)!
- OS differ considerably

File systems

In computing, a file system governs file organization and access. It controls how an operating system stores, organizes, and retrieves data on storage devices.

File systems differ between operating systems.

Files

- File = collection of data
- In statisitical computing: data, code, figures, tables, ...
- File names are used to distinguish files
- File names = name plus extension, separated by dot
- Example: figure.jpg
- Extension usually indicates file type
- Windows does not always show extensions!

Directories and drives

- Files are stored on drives and in directories
- Drive = (often) physical location, e.g., hard drive
- On Windows, drives have letters, e.g., C
- Directories = folders = cataloging system on a drive
- Directories have names, can have subdirectories
- Example: folder “Data” on drive C.

Paths

- Path = drive name + directory names + file name
- Path also includes separators
- Example:

```
C:\Files\Example\file.jpg
```

- Note the difference between a slash and a backslash

```
/ vs \
```

What does that mean for you?

- Organizing files well is key to handle complex projects
- Always make sure that you understand where you save a file so that you can find it again

Versions

- Software usually comes with a version number
- Typical format: Major - Minor - Bug fix
- Examples:
  - 1.3.2
  - 0.3.2
  - 5.13.1
- Applies to R, R packages, and lots of other software
- Could occasionally be important

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